Calculus I

Tutorial #12 (16–20 Dec 2024)

Topics:

- Alternating Series
- Power Series
- Taylor and Maclaurin Series

Problem 1. Determine if the alternating series converges or diverges. Some of the series do not satisfy the conditions of the Alternating Series Test.

(a) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{n}{10}\right)^n$.

(b) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{3\sqrt{n+1}}{\sqrt{n}+1}$.

Problem 2. Which of the series converge absolutely, which converge, and which diverge? Justify your answers.

(a) $\displaystyle\sum_{n=1}^{\infty}\frac{(-2)^{n+1}}{n+5^n}$.

(b) $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\ln n}{n-\ln n}$.

Problem 3. For each series, find the radius and interval of convergence. For what values of x does the series converge absolutely? conditionally?

(a) $\displaystyle\sum_{n=1}^{\infty}\frac{(x-1)^n}{\sqrt{n}}$.

(b) $\displaystyle\sum_{n=1}^{\infty}n^n x^n$.

Problem 4. Find the Taylor polynomials of orders 0, 1, 2, and 3 generated by $f(x) = \sin x$ at $a = \dfrac{\pi}{4}$.

Problem 5. Use power series operations to find the Taylor series at $x = 0$ for the functions $f(x) = e^x + \dfrac{1}{1+x}$.

Problem 6. Estimate the error if $P_3(x) = x - \dfrac{x^3}{6}$ is used to estimate the value of $\sin x$ at $x = 0.1$.

Problem 7. The estimate $\sqrt{1+x} = 1 + \dfrac{x}{2}$ is used when x is small. Estimate the error when $|x| < 0.01$.

Problem 8. Replace x by $-x$ in the Taylor series for $\ln(1+x)$ to obtain a series for $\ln(1-x)$. Then subtract this from the Taylor series for $\ln(1+x)$ to show that for $|x| < 1$,

$$\ln \frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Problem 9. How many terms of the Taylor series for $\ln(1+x)$ should you add to be sure of calculating $\ln(1.1)$ with an error of magnitude less than 10^{-8}? Justify your answer.

Problem 10. Consider the function $f(x) = \cos x$ near $x = 69$. What Taylor series would you choose to represent the function? (There may be more than one good answer.) Write out the first four nonzero terms of the series you choose.

THE END

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Answer to selected problems

1. (a) diverges (b) diverges
2. (a) absolutely converges (b) converges not absolutely
3. (a) converges absolutely $0 < x < 2$, converges conditionally $x = 0$
   (b) only converges (absolutely) $x = 0$
6. $|R_3(0.1)| \leq 42 \times 10^{-8}$
9. 8